

Literature Status Note: The Heisenberg Ball L_1 Distortion Exponent

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Status

Status: literature_already_answered. The open exponent problem in Austin–Naor–Tessera, *Sharp quantitative nonembeddability of the Heisenberg group into superreflexive Banach spaces*, arXiv:1007.4238, is answered by Naor–Young, *Foliated corona decompositions*, arXiv:2004.12522.

Source Question

Let B_R be the radius R word ball in the discrete Heisenberg group

$$\mathbb{H} = \langle a, b \mid aba^{-1}b^{-1} \text{ is central} \rangle,$$

and let $c_{L_1}(B_R)$ denote the least bi-Lipschitz distortion of (B_R, d_W) into L_1 .

In the subsection “Bi-Lipschitz distortion of balls”, Austin–Naor–Tessera recall the Cheeger–Kleiner–Naor estimate

$$c_{L_1}(B_R) \gtrsim (\log R)^c$$

for some universal $c > 0$, and state that evaluating the supremum over those $c > 0$ remains an important open problem.

Supporting Answer

The supporting paper is Assaf Naor and Robert Young, *Foliated corona decompositions*, arXiv:2004.12522. Its introduction states that the paper answers the asymptotic evaluation question for $c_{\ell_1}(B_n)$ and that the previously best lower bound was the Cheeger–Kleiner–Naor bound $c_{\ell_1}(B_n) \geq (\log n)^\delta$ for a small universal $\delta > 0$.

The decisive statement is Theorem 1.6:

$$c_{\ell_1}(B_n) \asymp \sqrt[4]{\log n} \quad \text{for every integer } n \geq 2.$$

Here B_n is the word ball in the discrete 3-dimensional Heisenberg group.

Conclusion

The source question asks for the supremum of exponents c such that

$$c_{L_1}(B_R) \gtrsim (\log R)^c.$$

Naor–Young’s two-sided estimate gives both a lower bound with $c = 1/4$ and an upper bound that rules out every $c > 1/4$. Hence the supremum is exactly

$$\boxed{\frac{1}{4}}.$$

This is a literature answer rather than a new result from this run. The supporting paper explicitly identifies the same asymptotic evaluation problem in the Cheeger–Kleiner–Naor line and resolves it.

Scope

This note concerns the 3-dimensional discrete Heisenberg group from the source paper. Higher-dimensional Heisenberg groups behave differently for L_1 : Naor–Young’s earlier vertical-versus-horizontal theorem gives order $\sqrt{\log n}$ for the 5-dimensional case. That does not change the answer to the source question because the source group is the 3-dimensional $\mathbb{H}$.

References

- [1] T. Austin, A. Naor, and R. Tessera, *Sharp quantitative nonembeddability of the Heisenberg group into superreflexive Banach spaces*, Groups Geom. Dyn. 7 (2013), 497–522; arXiv:1007.4238.
- [2] A. Naor and R. Young, *Foliated corona decompositions*, arXiv:2004.12522.
- [3] J. Cheeger, B. Kleiner, and A. Naor, *Compression bounds for Lipschitz maps from the Heisenberg group to L_1* , Acta Math. 207 (2011), 291–373.